

From OpenAI to Open Source AI

Navigating between Commercial Ownership and Collaborative Openness

WORLDLINE

24 October 2024

OCX 24

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Who are we?

We design payments technology that powers the growth of millions of businesses around the world.



7000+ engineers
in over 40 countries



Managing 43+ billion
transactions per year



€250M spent in R&D
every year



Handling 150+
payment methods

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The **early days** of LLMs

From rule-based and simpler statistical models to LLMs



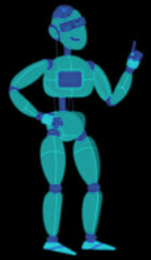
2010’s

Word embeddings
Word2Vec, GloVe



2017-2018

“*Attention is All You Need*"
Transformers



2020’s

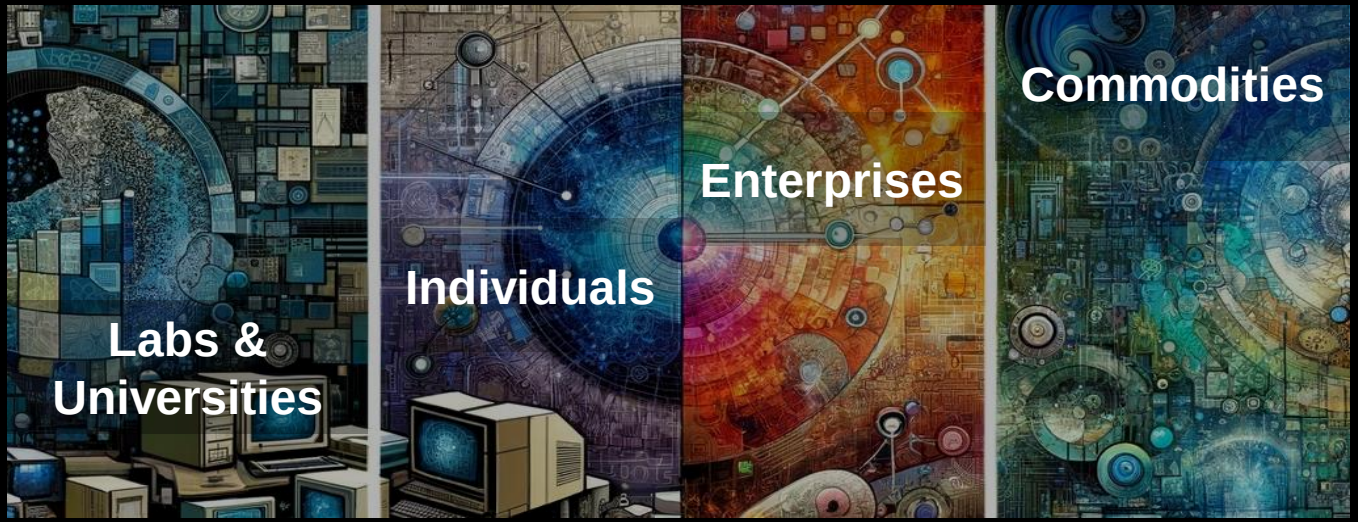
GenAI, ChatGPT
Responsibility concerns



Tomorrow?

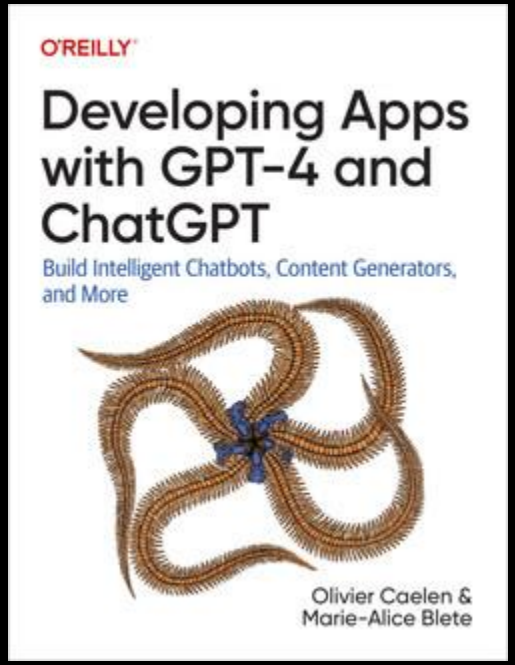
Small Language Models
Mobile, Agents & LAMs

GenAI is having its **Linux Moment**

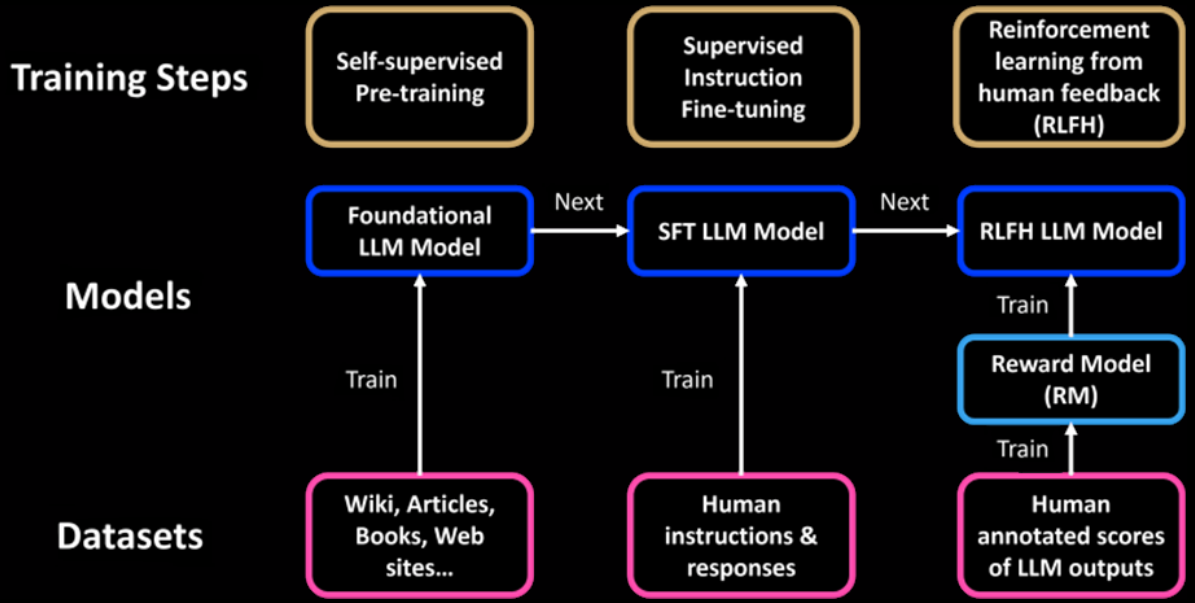


- Just like open source (and Internet), bust much faster!
- Dynamics between collaborative openness and commercial ownership
- Need of clarity on licenses

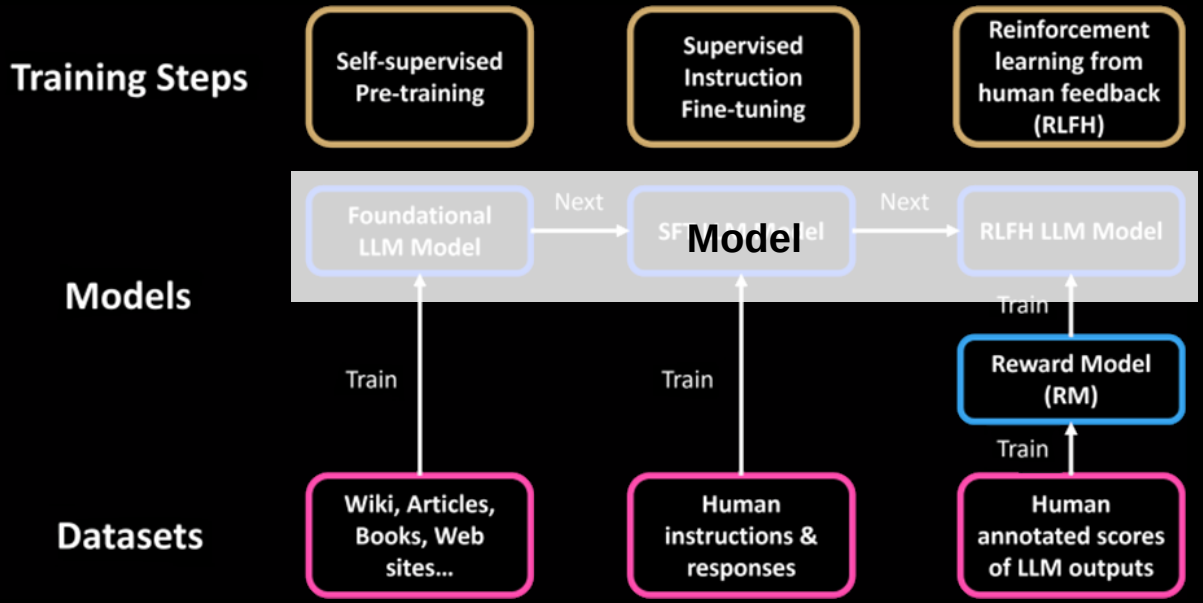
Defining **Openness** of a Model



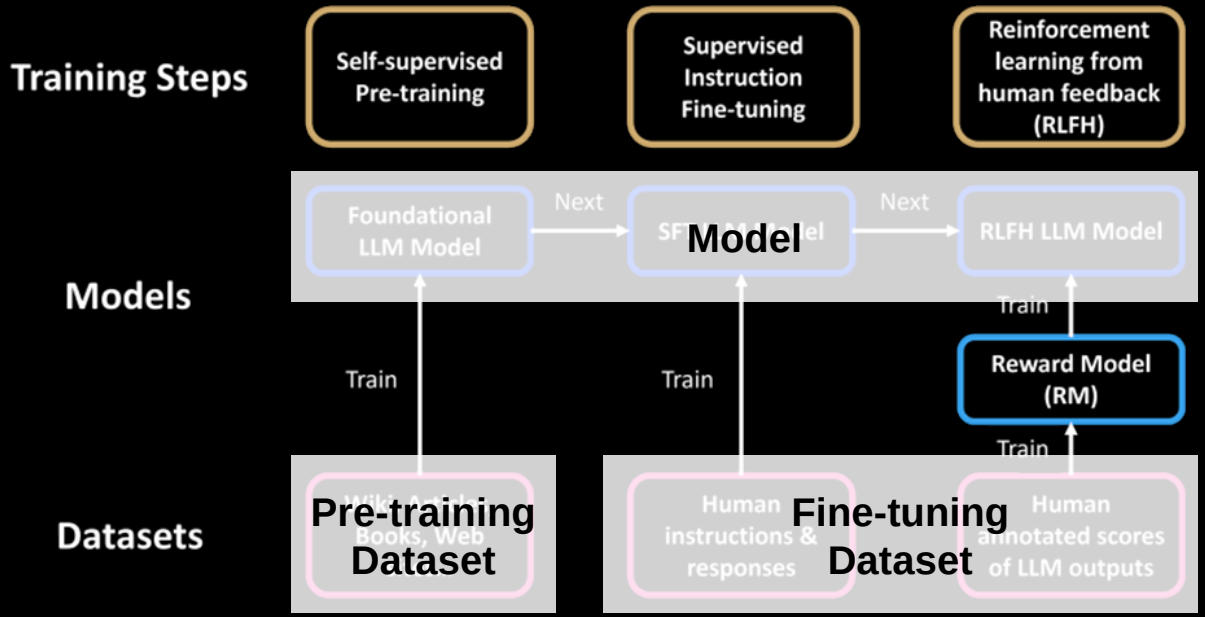
Defining **Openness** of a Model



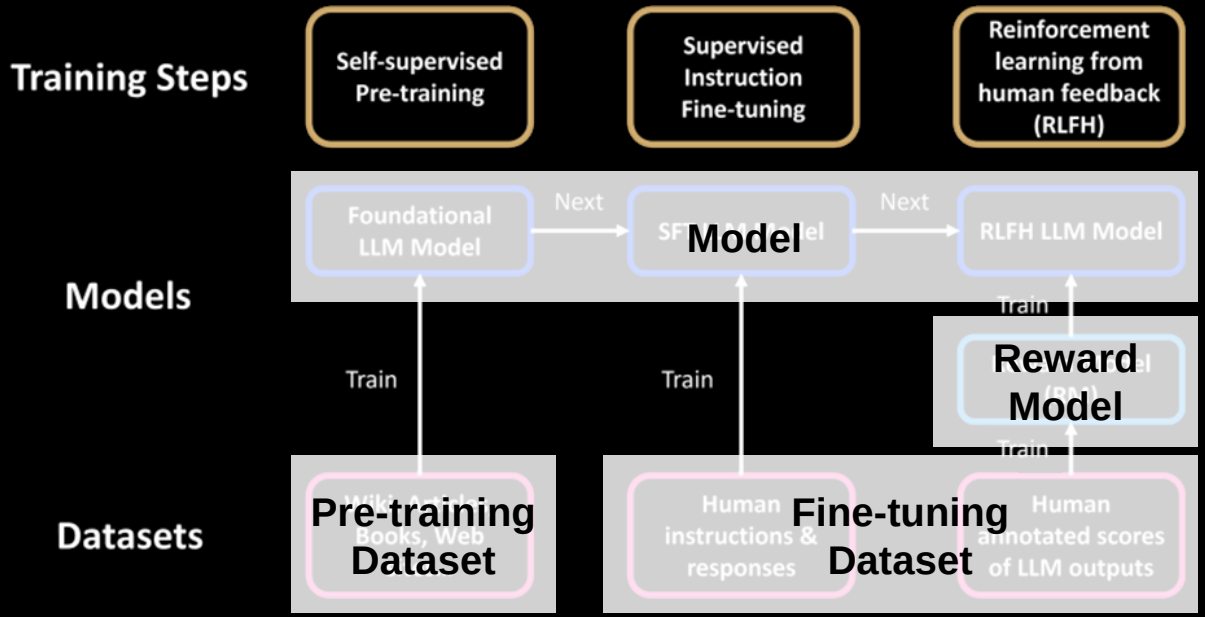
Defining **Openness** of a Model



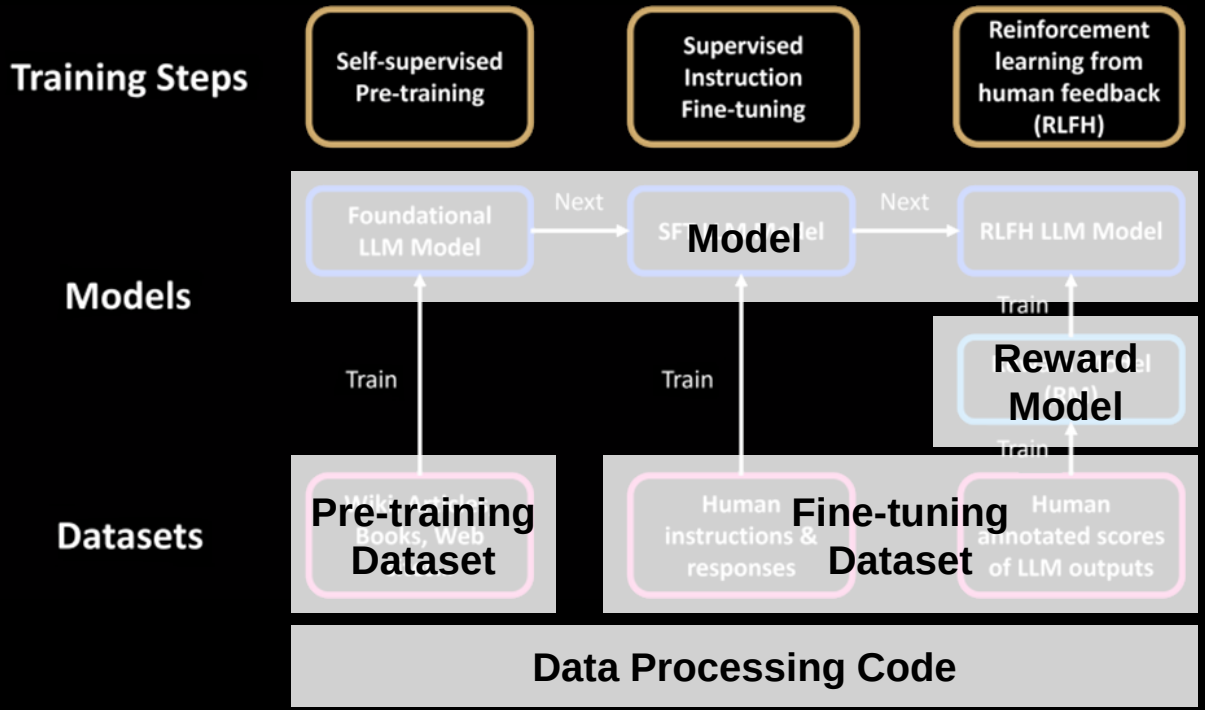
Defining **Openness** of a Model



Defining **Openness** of a Model



Defining **Openness** of a Model



Defining **Openness** of a Model

Score	Level	Description	Model (weights)	Pre-training Dataset	Fine-tuning Dataset	Reward model	Data Processing Code
0	**Closed**	**No access** to any public information, data or asset					
1	**Published research only**	**Research papers(s)** published but with no more information, data or asset					
2	**Restricted access**	Access to asset is possible only with **special agreement** (commercial, research…)					
3	**Open with limitations**	Access and reuse of asset is possible but with certain **limitations on usage**					
4	**Totally open**	Access and reuse of asset is possible **without restriction on usage** (ex. open source license)					

Market-Leading Player: OpenAI

Deviation from original vision of research transparency & openness



GPT-1 & 2

Non/For-profit (US)		
Component	**Score**	**Level description**
Model	4	Totally open
Dataset	1	Published research only
Code	1	Published research only

→

GPT-3.x & 4.x/o

0	Closed

ChatGPT

research paper only

Market-Leading Player: OpenAI

Deviation from original vision of research transparency & openness



GPT-1 & 2

		Non/For-profit (US)
		only
Code	1	Published research only

> *You may not: […] Use **Output** to develop models that compete with OpenAI.*

GPT-3.x & 4.x/o

0	Closed

ChatGPT

research paper only



No training of other commercial LLMs

Market-Leading Player: Google

Transition from open research to a pragmatic approach

Google BERT

Enterprise (US)		
Component	Score	Level description
Model	4	Totally open
Dataset	2	Restricted access
Code	4	Totally open

→

PaLM 2

Gemini

1	Published research only
1	Published research only
0	Closed

↔

Gemma

3	Open with limitations
1	Published research only
4	Toolchain available

Market-Leading Player: Google

Transition from open research to a pragmatic approach

Google BERT

PaLM 2

Gemma

	Gemma
Component	3 Open with limitations
Model	1 Published research only
Dataset	
Code	4 Totally open / 0 Closed / 4 Toolchain available

*You may **not use nor allow others to use** Gemma or Model Derivatives to: [illegals activities, unlicensed practices of profession, abuse, security bypass and promotion of hatred, abuse, violence, monitoring people without consent, misinformation/defamation, automate decisions concerning human rights and well-being, etc.]*

Responsible AI contradicts Open Source Definition

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Other **Big Players**

Catching up and making their mark in the GenAI Gold Rush

Create their own (open) models













Partner for Infrastructure (inference and training)

Market-Leading Player: Meta

Journey to openness

RoBERTa

Enterprise (US)		
Component	**Score**	**Level description**
Model	4	Totally open
Dataset	3	Open with limitations
Code	4	Totally open

LLaMA by Meta

3	Open with limitations
1	Published research only
1	Published research only

Market-Leading Player: Meta

Journey to openness

RoBERTa

LLaMA

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Restriction on usage: license for platforms with 700+ M users

Market-Leading Player: Meta

Journey to openness

RoBERTa

LLaMA by Meta

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LLaMA 3 now more restrictive on **redistribution and reuse**

Llama 2 offspring's: Alpaca and Vicuna

Fine-tuned models from Llama 2 by universities





Research (US)		
Component	Score	Level description
Model	3	Open with limitations
Pre-training Dataset	1	Published research only
Fine-tuning Dataset	2	Research use only
Code	4	Under Apache 2 license



Restrictions from **both** Llama 2 and OpenAI (ShareGPT)

Collaborative **foundational** LLMs

	EleutherAI				BLOOM				MISTRAL AI_	
	Non-profit (US)		Research (UAE)		Research (EU)		Research (US)		Enterprise (FR)	
	EleutherAI GPT-J		Falcon		BLOOM		OpenLLaMa		Mistral	
Model	4	Access and reuse without restriction	3	Open with limitations	3	Open RAIL license	4	Access and reuse without restriction	4	Access and reuse without restriction
Dataset	3	Open with limitations	4	Access and reuse without restriction	3	Open with limitations	4	Access and reuse without restriction	0	No public information or access
Code	4	Completely open	1	General instructions	4	Completely open	1	Just examples	4	Completely open

Dataset fuzziness: *please refer to the specific license depending on the subset you use*

Notion of **responsible** usage

Collaborative **foundational** LLMs

	Non-profit (US)	Research (UAE)	Research (EU)	Research (US)	Enterprise (FR)
	EleutherAI GPT-J	Falcon	BLOOM	OpenLLaMa	Mistral
Model	4 open	1 instructions	4 open	1 examples	4 open
Dataset					
Code					

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Modified open source licenses

Mistral AI's French sauce

Navigating both open and close waters

Mix of AI Models

- Mixture-of-Experts (SMoE): Mixtral 8x7B, 8x22B
- Foundational and fine-tuned models (i.e. Mathstral)
- Small Language Models: Ministral 3B & 8B

Mix of Business Models & Licenses

- "Open Source" models, *mistral-finetune* SDK
- Commercial: optimized Small, Large & Embed Models
- Sustainable openness: new non-production license for *codestral*



Just like with Open Source, rise of **Community VS Enterprise**

Mistral AI's French sauce

Navigating both open and close waters

Mix

M

F

Mix

"C

Commercial: optimized Small, Large & Embed Models

Sustainable openness: new non-production license for codestral

> ***MNPL - 3.2. Usage Limitation***
> *- You shall only use the Mistral Models and Derivatives (whether or not created by Mistral AI) for testing, research, Personal, or evaluation purposes in Non-Production Environments;*
> *- Subject to the foregoing, You shall not supply the Mistral Models or Derivatives in the course of a commercial activity, whether in return for payment or free of charge, in any medium or form, including but not limited to through a hosted or managed service (e.g. SaaS, cloud instances, etc.), or behind a software layer.*



Just like with Open Source, revisiting ***Open*** **in Cloud era**

Collaborative **fine-tuned** LLMs

	Dolly		BLOOMChat		Zephyr		LLM360		OLMo-Instruct	
	Enterprise (US)		Enterprise (US)		Enterprise (US)		Consortium (UAE/US)		Research (US)	
	Dolly		BLOOMChat		Zephyr		LLM360		OLMo-Instruct	
Model	4	Based on GPT-J	3	Based on BLOOM	4	Based on Mistral	4	Open source	4	Open source
Pre-training Dataset	3	Based on GPT-J	3	Based on BLOOM	0	Based on Mistral	4	RedPajama, Falcon, StarCoder	3	Dolma (ImpACT MR)
Fine-tuning Dataset	4	Access and reuse without restriction	4	Dolly and LAION	2	Research use only (OpenAI)	2	Research use only (OpenAI)	3	Tülu 2 (IMPACT LR)
Reward model	0	No public information available	0	No public information available	3	Paper and code examples	0	No public information available	4	UltraFeedback (MIT)
Code	4	Open source	3	OpenRAIL	3	Example code available	4	Open source	4	Open source

Impact of foundational model or pre-training datasets

Collaborative **fine-tuned** LLMs

	Dolly	BLOOMChat	Zephyr	LLM360	OLMo
	Enterprise (US)	Enterprise (US)	Enterprise (US)	Consortium (UAE/US)	Research (US)
Model					
Pre-trainin Dataset					MR)
Fine-tunin Dataset					LR)
Reward model					dback
Code					

AI2 ImpACT Licenses - Restrictions

[…] a. military weapons purposes […]

b. purposes of military surveillance […]

c. purposes of generating or disseminating information or content […] without expressly and intelligibly disclaiming that the text is machine generated;

d. purposes of 'real time' remote biometric processing […]

e. fully automated decision-making without a human in the loop […] as spreading misinformation […]

f. purposes of the predictive administration of justice, law enforcement, immigration, or asylum processes, such as predicting an individual will commit fraud/crime

Responsible AI contradicts Open Source Definition

Other aspects of GenAI's **Linux Moment**

Democratize and Decentralize (re)use and innovation

Other aspects of GenAI's **Linux Moment**

Democratize and Decentralize (re)use and innovation

Collaborative Tools & Ecosystems





Notebooks
Communities
New Business Models

Other aspects of GenAI's **Linux Moment**

Democratize and Decentralize (re)use and innovation

Collaborative Tools & Ecosystems





Notebooks
Communities
New Business Models

Hardware Optimization





AI Chips
Quantization
Decentralization

Other aspects of GenAI's **Linux Moment**

Democratize and Decentralize (re)use and innovation

Collaborative Tools & Ecosystems





Notebooks
Communities
New Business Models

Hardware Optimization





AI Chips
Quantization
Decentralization

Opensource Tools & Frameworks





Do One Thing Well
Interoperable Standards
Beyond Python

Key **takeaways**



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Key **takeaways**



- **Closed APIs → Open Weights → Free AI (as in freedom)**
  - Datasets and upstream transitivity
  - Competitive clauses
  - Responsible AI restrictions

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- **Closed APIs → Open Weights → Free AI (as in freedom)**
  - Datasets and upstream transitivity
  - Competitive clauses
  - Responsible AI restrictions
- **Open Research → Competitive Market → Coopetitive Ecosystem**
  - Openness fosters reuse and collaboration
  - Collaboration brings commoditization and innovation

Key **takeaways**



- **Closed APIs → Open Weights → Free AI (as in freedom)**
  - Datasets and upstream transitivity
  - Competitive clauses
  - Responsible AI restrictions
- **Open Research → Competitive Market → Coopetitive Ecosystem**
  - Openness fosters reuse and collaboration
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Just like Open Source!



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Thank you!